

SnapBurst Product Overview

Governed Autonomy for Unmanned Aircraft

SnapBurst is a governed-autonomy product for unmanned aircraft.

It operates between mission autonomy and aircraft-command execution, controlling which proposed actions are permitted to reach the flight controller.

Autonomy proposes. SnapBurst governs. The flight controller executes permitted actions.

SnapBurst is designed for operations in which communications, command authority, aircraft state and mission conditions may change during flight. It ensures that aircraft behaviour remains authorised, bounded, attributable and auditable through those changes.

The product structure is:

- **SnapBurst** is the standalone governed-autonomy product for unmanned-aircraft operators and system integrators.
- **SnapBurstOS** is the licensed integration form through which aircraft manufacturers, autonomy developers and fleet-platform providers can incorporate SnapBurst capability into their own drone products.
- **SnapSpace OS** is the proprietary deterministic governance core beneath both forms.

SnapBurst and SnapBurstOS are specific to unmanned-aircraft applications. Neither provides unrestricted access to the underlying SnapSpace OS technology or cross-industry rights to that governance core.

The Operational Problem

BVLOS and autonomous aircraft cannot assume that command-and-control connectivity will remain continuously available.

When a C2 link degrades or disappears, conventional aircraft systems commonly hold position, return home, land, abort the mission or wait for operator intervention.

Those responses may protect the aircraft, but they also interrupt useful operations.

The deeper problem is not simply loss of telemetry or connectivity. It is loss of clear command authority.

The aircraft system must still determine:

- which authority remains valid;
- what actions remain permitted;
- whether the current task may continue;
- how aircraft and mission state are preserved;
- how restored command is verified;
- how control is returned;
- and how the full event is evidenced afterward.

SnapBurst addresses that governance problem.

It provides a controlled decision layer between proposed aircraft behaviour and physical command execution.

What SnapBurst Does

In FreeFlight Recovery and Autonomous Work Mode, SnapBurst applies a deterministic governance decision before relevant aircraft commands are dispatched.

For each proposed action, it evaluates the available authority, command source, aircraft identity, mission state, operating envelope and supporting evidence.

It then produces one of three controlled aircraft-domain outcomes.

Permit

The proposed action is authorised under the current conditions and may proceed to execution.

Reject

The proposed action is outside the authorised conditions and is blocked.

Hold

The action cannot yet be conclusively permitted or rejected using the available authority and evidence.

It remains contained within the system pending additional evidence, remediation, a second review or human intervention.

Permit, Reject and Hold are SnapBurst's aircraft-domain governance outcomes. They are implemented through the underlying SnapSpace OS deterministic decision model.

Within the governed command path, SnapBurst preserves the relationship between:

- command origin;
- aircraft identity;
- active authority;
- mission state;
- aircraft state;
- operating conditions;
- governance outcome;
- dispatched action;
- aircraft response;
- and execution evidence.

This creates an attributable record of what was proposed, what was permitted, what was blocked, what was contained and what the aircraft actually did.

Three Operating Modes

Manual Flight

In Manual Flight, the human operator commands the aircraft through the approved C2 path.

SnapBurst correlates:

- the command source;
- aircraft identity;
- active authority;
- aircraft and mission state;
- the governance outcome;
- the command dispatched;
- and the resulting aircraft response.

The operator remains the active command authority while the approved command path remains valid.

In Manual Flight, SnapBurst correlates the command source, aircraft identity, authority state, command dispatch and aircraft response for attribution and evidence.

FreeFlight Recovery

FreeFlight Recovery applies when the approved external command path becomes unavailable.

The aircraft does not move into unrestricted autonomy.

Mission logic may continue proposing actions, but continuation is limited to the pre-authorised operating envelope.

SnapBurst evaluates each proposed action against the remaining authority, aircraft identity, current state, mission conditions and approved boundaries.

Actions inside the authorised envelope may be permitted.

Actions outside the authorised envelope are rejected or held and do not reach the flight controller.

When communications return, external command authority is not restored automatically. SnapBurst verifies the restored command condition, checks the current aircraft and mission state, and manages a governed handback.

The link disappears. The mission remains governed.

FreeFlight Recovery is bounded mission continuation during command-path interruption. It is not uncontrolled autonomy and it does not treat unavailable external authority as active control.

Autonomous Work Mode

Autonomous Work Mode supports approved tasks in which mission logic proposes aircraft actions while the operator retains visibility and intervention capability.

The mission system proposes the work.

SnapBurst governs what may proceed.

The flight controller executes only permitted actions.

Autonomous Work Mode remains bounded by the same authority, identity, aircraft-state, mission-state and evidence controls used across the SnapBurst operating model.

The autonomy system remains responsible for producing mission proposals. SnapBurst remains responsible for governing whether those proposals may reach aircraft-command execution.

Governed C2-Loss Sequence

A typical SnapBurst operating sequence is:

- 1 The aircraft begins inside an approved mission and operating envelope.
- 2 The authorised command source and operating authority are established.
- 3 Aircraft identity, mission state, aircraft state and authority context are maintained.
- 4 A C2 interruption is detected.
- 5 The unavailable external command path is no longer treated as active authority.
- 6 Mission logic may continue proposing actions inside the approved envelope.
- 7 SnapBurst evaluates each proposed action.
- 8 SnapBurst permits, rejects or holds the action.
- 9 Unauthorised or unresolved commands do not reach the flight controller.
- 10 Permitted actions are dispatched through the governed command path.
- 11 Aircraft responses are correlated with the decision and dispatched command.
- 12 Communications are restored.
- 13 The returning command source is verified against the current aircraft, mission and authority state.
- 14 External command returns only through a governed handback.
- 15 The complete decision, dispatch, response and transition sequence remains available as evidence.

FreeFlight is bounded mission continuation, not uncontrolled autonomy.

Core Product Capabilities

Command-Authority Governance

In governed modes, SnapBurst determines whether the active authority is valid for the proposed action, aircraft and operating state.

The presence of a communication link alone does not establish valid authority.

The absence of a link does not automatically remove every previously authorised mission condition.

SnapBurst evaluates the authority that actually remains applicable at the time of the proposed action.

Authorised-Envelope Enforcement

Autonomous proposals and FreeFlight Recovery actions are evaluated against approved operating conditions before execution.

The authorised envelope may include:

- aircraft identity;
- mission identity;
- permitted task;
- operating area;
- command source;
- current mode;
- aircraft state;
- mission state;
- communications condition;
- and other defined operating constraints.

Proposals outside the authorised envelope are rejected or held before they reach the flight controller.

Governed FreeFlight

Aircraft operations may continue through simulated C2 interruption without treating missing command authority as active control.

Continuation is limited to actions that remain authorised under the approved mission and operating envelope.

Controlled Restoration and Handback

Restored communications do not automatically override the aircraft's current state.

The returning command source must be verified.

The current aircraft state, mission state and active authority must be reconciled before external control is restored.

Authority is returned through a governed transition rather than an uncontrolled switch.

State Continuity

Aircraft, mission, authority and governance state are retained across command-loss and restoration events.

This allows SnapBurst to evaluate proposed actions against the state that actually exists, rather than treating communications restoration as a return to an earlier or assumed condition.

Multi-Aircraft Isolation

Each aircraft lane maintains its own:

- identity;
- mission state;
- aircraft state;
- command source;
- authority;
- governance decision;
- dispatch path;
- aircraft response;
- and evidence record.

A condition affecting one aircraft must not improperly alter another aircraft's authority, state, command path or governance result.

Decision Evidence

Governance decisions, state transitions, commands and aircraft responses are correlated for:

- testing;
- operational assurance;
- regulatory engagement;
- programme review;
- incident analysis;
- post-operation investigation;
- and future certification evidence.

The evidence record distinguishes between what was proposed, what was authorised, what was dispatched and what the aircraft acknowledged or executed.

Position Within the Aircraft System

SnapBurst does not replace the aircraft platform.

Existing aircraft and OEM systems continue to provide:

- the airframe;
- propulsion;
- the flight controller;
- communications;
- ground-control interfaces;
- mission planning;
- mission autonomy;
- navigation;
- perception;
- detect-and-avoid capability;
- sensors;
- payload systems;
- and aircraft-specific safety functions.

SnapBurst adds:

- command-authority governance;
- authorised-envelope enforcement;
- C2-loss handling;
- bounded FreeFlight continuation;
- governed command restoration;
- controlled handback;
- aircraft-lane attribution;
- deterministic governance outcomes;
- and correlated decision evidence.

The operating relationship is:

****Sensors detect.**

Mission autonomy proposes.

SnapBurst governs.

The flight controller executes permitted actions.

Evidence records what occurred.**

SnapBurst may govern an avoidance action proposed by an external perception or detect-and-avoid system.

It does not claim to detect the obstacle, classify the hazard or calculate the avoidance path itself.

Those functions remain with the relevant perception, autonomy or detect-and-avoid system.

SnapBurst governs whether the resulting proposed aircraft action is authorised to proceed.

Integration Position

The current implementation and validation environment uses PX4 and ROS 2.

SnapBurst is intended to integrate with existing aircraft, autonomy and fleet systems rather than requiring those systems to be replaced.

A bounded integration would typically connect:

- aircraft identity;
- aircraft state;
- mission identity;
- mission state;
- command-source information;
- authority information;
- mission-command inputs;
- autonomous action proposals;
- communications state;
- flight-controller command interfaces;
- aircraft acknowledgements;
- and aircraft responses.

Within the governed autonomy path, SnapBurst applies governance at the aircraft-command boundary.

It determines whether a proposed action may proceed and creates a correlated evidence record linking the proposal, authority, governance outcome, command dispatch and aircraft response.

Within the current PX4 implementation, the operating relationship is:

Autonomy proposes. SnapBurst governs. PX4 executes permitted actions.

The same governance model is intended to be adapted to other aircraft, autonomy and fleet architectures through bounded integration interfaces.

Current Validation Position

SnapBurst has been exercised in PX4 and ROS 2 software-in-the-loop environments.

The current evidence supports:

- governed command decisions before aircraft-command dispatch;
- Permit, Reject and Hold governance outcomes;
- controlled operation through simulated C2 interruption;
- bounded FreeFlight continuation;
- prevention of unauthorised command dispatch;
- governed restoration of external command;
- controlled handback;
- aircraft-state continuity;
- mission-state continuity;
- authority-state continuity;
- independent multi-aircraft attribution;
- independent multi-aircraft isolation;
- repeated scenario execution;
- and correlated aircraft-command acknowledgement.

The validation work demonstrates the current software behaviour inside the tested simulation environment.

Detailed test results, execution campaigns, scenario outcomes and numerical evidence are maintained separately in the **SnapBurst Validated Evidence Summary**.

That evidence document is the authoritative location for specific test counts, execution durations, aircraft counts and validation results.

Outside Current Claim Boundary

The current evidence does not establish:

- certified BVLOS operation;
- regulatory approval for operational deployment;
- physical-flight performance;
- hardware-in-the-loop performance;
- aircraft-specific integration performance;
- operational performance on a third-party aircraft;
- detect-and-avoid performance;
- obstacle detection;
- avoidance-path generation;
- distributed swarm coordination;
- or unrestricted autonomous operation.

These remain outside the current validated claim boundary until supported by the appropriate integration, physical-flight, operational, regulatory or certification evidence.

Commercial Product Structure

SnapBurst is intended to reach the unmanned-aircraft market through two related but distinct commercial routes.

Both routes use the same underlying governed-autonomy capability, but they serve different customers and deployment models.

SnapBurst Standalone Product

SnapBurst is the standalone governed-autonomy product.

It is intended for unmanned-aircraft operators, programme owners and system integrators that need to add governed command execution, C2-loss handling, bounded mission continuation, controlled restoration and evidence to an existing aircraft operation.

The standalone product can be integrated around an existing combination of:

- aircraft platform;
- flight controller;
- mission system;
- autonomy system;
- communications system;
- fleet-management system;
- and operational-control environment.

The customer continues using its existing aircraft and autonomy technology.

SnapBurst adds the governance boundary between proposed actions and aircraft-command execution.

A standalone SnapBurst deployment may support:

- operator-specific integration;
- system-integration projects;
- pilot programmes;
- regulated operational trials;
- supervised BVLOS development;
- aircraft-fleet integration;
- assurance evidence generation;
- and operational governance around existing autonomous systems.

The standalone product remains identifiable as SnapBurst within the customer's operating architecture.

SnapBurstOS Integration and Licensing

SnapBurstOS is the third-party drone-industry integration and licensing form of the SnapBurst capability.

It is intended for:

- aircraft manufacturers;
- drone OEMs;
- autopilot and flight-stack providers;
- autonomy-system developers;
- fleet-platform companies;

- mission-platform providers;
- and other unmanned-aircraft technology companies.

SnapBurstOS allows those companies to incorporate SnapBurst governance into their own aircraft, autonomy or fleet products.

This route is not simply a standalone SnapBurst installation operated alongside the customer's product.

It is the licensed product form for deeper third-party integration, embedding or commercial incorporation within the partner's own unmanned-aircraft offering.

Depending on the agreed integration, SnapBurstOS may be incorporated at:

- the aircraft-command boundary;
- the autonomy-to-flight-controller interface;
- the fleet-command layer;
- the mission-execution layer;
- the aircraft supervisory layer;
- or another defined governance point within the partner architecture.

A SnapBurstOS licence may support:

- OEM product integration;
- embedded governed-autonomy capability;
- licensed software components;
- product-specific adaptation;
- aircraft-family integration;
- fleet-platform integration;
- partner-controlled deployment;
- and commercial distribution as part of a third-party drone product.

The third party retains responsibility for its aircraft, autonomy, mission logic, communications, operational approvals and product-specific safety case.

SnapBurstOS provides the licensed governance capability within that wider product.

SnapBurstOS remains specific to unmanned-aircraft applications.

A SnapBurstOS licence does not provide:

- unrestricted access to SnapSpace OS;
- ownership of the underlying deterministic governance core;
- cross-industry licensing rights;
- rights to apply SnapSpace OS outside the agreed unmanned-aircraft scope;
- or authority to redistribute the underlying core independently of the licensed SnapBurstOS product.

SnapSpace OS Relationship

SnapSpace OS is the proprietary deterministic governance core beneath SnapBurst and SnapBurstOS.

It provides the underlying decision model and governance mechanisms from which the aircraft-specific products are constructed.

The commercial relationship is therefore:

SnapSpace OS

Proprietary cross-domain governance core.

SnapBurst

Standalone governed-autonomy product for unmanned-aircraft operators and system integrators.

SnapBurstOS

Licensed third-party drone-industry integration form for manufacturers, autonomy developers and fleet-platform providers.

This structure allows SnapBurst capability to be sold directly as a standalone product while also supporting deeper licensed incorporation into third-party unmanned-aircraft products without exposing or transferring unrestricted rights to the underlying SnapSpace OS technology.

Partner and Programme Collaboration

SnapBurst can support bounded collaboration with:

- aircraft manufacturers;
- drone operators;
- autonomy developers;
- fleet-platform providers;
- communications providers;
- test facilities;
- regulators;
- research organisations;
- defence and aerospace programmes;
- accelerators;
- and innovation partners.

A typical collaboration may combine:

- a partner aircraft or aircraft model;
- an existing flight controller;
- an existing mission or autonomy system;
- a defined command-loss scenario;
- a bounded operating envelope;
- SnapBurst governance integration;
- simulated or physical test execution;
- and a jointly reviewed evidence package.

The purpose of such collaboration is not to replace the partner's existing aircraft technology.

It is to establish whether governed-autonomy capability can be integrated at the command boundary and evidenced under defined operational conditions.

Potential collaboration routes include:

- standalone SnapBurst trials;
- system-integration projects;
- supervised BVLOS development programmes;
- aircraft-specific validation;
- SnapBurstOS licensing evaluations;
- OEM integration studies;
- fleet-platform integration;
- regulatory innovation programmes;
- and jointly funded research or demonstration activity.

Product Value

SnapBurst addresses the operational trust gap between autonomous decisions and physical aircraft actions.

Its value is not simply that an aircraft can continue operating when communications or command conditions change.

Its value is that continued operation remains:

- authorised;
- bounded;
- attributable;
- isolated by aircraft;
- recoverable;
- reviewable;
- and auditable.

SnapBurst provides a controlled path from supervised aircraft operation toward governed autonomous operation without replacing the aircraft, flight controller, communications system, mission-autonomy platform or detect-and-avoid system.

For operators and system integrators, SnapBurst provides a standalone governance product around existing aircraft operations.

For aircraft manufacturers, autonomy developers and fleet-platform providers, SnapBurstOS provides a licensed route for incorporating that capability into third-party drone products.

In both cases, the underlying SnapSpace OS governance core remains proprietary and controlled.

SnapBurst governs the mission when command authority becomes uncertain—and preserves the evidence of what happened afterward.